4. Practical Exercise: Build and Run a Java Application with Maven, Migrate the Same Application to Gradle.

Step 1: Creating a Maven Project

- To create a basic Maven project using the command line, you can use the following command:

  mvn archetype:generate -DgroupId=com.example -DartifactId=maven-example -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false

Step 2: Open the pom.xml File

- You can manually navigate the **project folder** named call **maven-example** and open the file pom.xml and copy the below code and paste it then save it.
- In case if you not getting project folder then type command in your cmd.
  - **cd maven-example** – is use to navigate the project folder.
  - **notepad pom.xml** – is use to open pom file in notepad after that type the below code in POM.xml file

```
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0"
        xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
        xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
http://maven.apache.org/xsd/maven-4.0.0.xsd">
   <modelVersion>4.0.0</modelVersion>
   <groupId>com.example</groupId>
   <artifactId>maven-example</artifactId>
   <version>1.0-SNAPSHOT</version>
   <dependencies>
      <dependency>
         <groupId>junit</groupId>
         <artifactId>junit</artifactId>
         <version>4.12</version>
         <scope>test</scope>
      </dependency>
   </dependencies>
   <build>
      <plugins>
         <plugin>
            <groupId>org.apache.maven.plugins</groupId>
            <artifactId>maven-compiler-plugin</artifactId>
            <version>3.8.1</version>
            <configuration>
               <source>1.8</source>
               <target>1.8</target>
            </configuration>
         </plugin>
```

```
        </plugins>
    </build>
</project>
```

Step 3: Open Java Code (App.java) File

- Open a file **App.java** inside the **src/main/java/com/example/** directory.
- After opening the **App.java** copy the below code and paste it in that file then save it.

```
package com.example;
public class App {
    public static void main(String[] args) {
        System.out.println("Hello, Maven");
        System.out.println("This is the simple realworld example....");
        int a = 5;
        int b = 10;
        System.out.println("Sum of " + a + " and " + b + " is " + sum(a, b));
    }
    public static int sum(int x, int y) {
        return x + y;
    }
}
```

Note: before building the project make sure you are in the project folder if not navigate the project folder type command in your command prompt **cd maven-example**.

Step 4: Run the Project

To build and run this project, follow these steps:

- Open the command prompt in the project directory and run the following command to build the project

```
mvn clean install
```

```
s\Admin\.m2\repository\com\example\maven-example\1.0-SNAPSHOT\maven-example-1.
-SNAPSHOT.pom
[INFO] Installing C:\Users\Admin\maven-example\maven-example\target\maven-exam
le-1.0-SNAPSHOT.jar to C:\Users\Admin\.m2\repository\com\example\maven-example
1.0-SNAPSHOT\maven-example-1.0-SNAPSHOT.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  02:42 min
[INFO] Finished at: 2025-03-04T19:29:06+05:30
[INFO] ------------------------------------------------------------------------
```

- Run the program with below command:

```
mvn exec:java -Dexec.mainClass="com.example.App"
```

```
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/
resolver/maven-resolver-api/1.4.1/maven-resolver-api-1.4.1.jar (149 kB at 93 kB
/s)
Hello, Maven
This is the simple realworld example....
Sum of 5 and 10 is 15
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  10.394 s
[INFO] Finished at: 2025-03-04T19:29:32+05:30
[INFO] ------------------------------------------------------------------------
```

Step 5: Migrate the Maven Project to Gradle

1. **Initialize Gradle**: Navigate to the project directory (**maven-example**) or **(gradle-example)** and run:

gradle init

- It will ask **Found a Maven build. Generate a Gradle build from this? (default: yes) [yes, no]**
  - Type **Yes**
- **Select build script DSL:**
  - 1: Kotlin
  - 2: Groovy
  - Enter selection (default: Kotlin) [1..2]
    - Type **2**
- **Generate build using new APIs and behavior (some features may change in the next minor release)? (default: no) [yes, no]**
  - Type **No**

```
C:\Users\Admin\maven-example\maven-example>gradle init
Starting a Gradle Daemon, 1 incompatible and 1 stopped Daemons could not be reu
sed, use --status for details

Found a Maven build. Generate a Gradle build from this? (default: yes) [yes, no
] yes

Select build script DSL:
  1: Kotlin
  2: Groovy
Enter selection (default: Kotlin) [1..2] 2

Generate build using new APIs and behavior (some features may change in the nex
t minor release)? (default: no) [yes, no] no


> Task :init
Maven to Gradle conversion is an incubating feature.
For more information, please refer to https://docs.gradle.org/8.12.1/userguide/
migrating_from_maven.html in the Gradle documentation.

BUILD SUCCESSFUL in 2m 7s
1 actionable task: 1 executed
```

2. **Navigate the project folder and open build.gradle file then add the below code and save it.**

```
plugins {
    id 'java'
}
group = 'com.example'
version = '1.0-SNAPSHOT'
repositories {
    mavenCentral()
}

dependencies {
    testImplementation 'junit:junit:4.12'
}

task run(type: JavaExec) {
    main = 'com.example.App'
    classpath = sourceSets.main.runtimeClasspath
}
```

Step 6: Run the Gradle Project

- **Build the Project**: In the project directory (gradle-example or maven-example), run the below command to build the project:

gradlew build

```
C:\Users\Admin\maven-example\maven-example>gradlew build
Downloading https://services.gradle.org/distributions/gradle-8.12.1-bin.zip
.............10%.............20%.............30%.............40%.............50
%.............60%.............70%.............80%.............90%.............1
00%
Calculating task graph as no cached configuration is available for tasks: buil

[Incubating] Problems report is available at: file:///C:/Users/Admin/maven-exam
ple/maven-example/build/reports/problems/problems-report.html

Deprecated Gradle features were used in this build, making it incompatible with
 Gradle 9.0.

You can use '--warning-mode all' to show the individual deprecation warnings an
d determine if they come from your own scripts or plugins.

For more on this, please refer to https://docs.gradle.org/8.12.1/userguide/comm
and_line_interface.html#sec:command_line_warnings in the Gradle documentation.

BUILD SUCCESSFUL in 8m 17s
4 actionable tasks: 4 executed
Configuration cache entry stored.
```

- **Run the Application**: Once the build is successful, run the application using below command:

gradlew run

```
C:\Users\Admin\maven-example\maven-example>gradlew run
Calculating task graph as no cached configuration is available for tasks: run

> Task :run
Hello, Maven
This is the simple realworld example....
Sum of 5 and 10 is 15

[Incubating] Problems report is available at: file:///C:/Users/Admin/maven-exam
ple/maven-example/build/reports/problems/problems-report.html

Deprecated Gradle features were used in this build, making it incompatible with
 Gradle 9.0.

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d determine if they come from your own scripts or plugins.

For more on this, please refer to https://docs.gradle.org/8.12.1/userguide/comm
and_line_interface.html#sec:command_line_warnings in the Gradle documentation.

BUILD SUCCESSFUL in 4s
2 actionable tasks: 1 executed, 1 up-to-date
Configuration cache entry stored.
```

Step 7: Verify the Migration

- **Compare the Output:** Make sure that both the **Maven** and **Gradle** builds produce the same output:

  - **Maven Output**:

```
Hello, Maven
This is the simple realworld example....
Sum of 5 and 10 is 15
```

- **Gradle Output**:

```
Hello, Maven
This is the simple realworld example....
Sum of 5 and 10 is 15
```